Journal of Colloid and Interface Science 623 (2022) 870–882

Contents lists available at ScienceDirect

Journal of Colloid and Interface Science

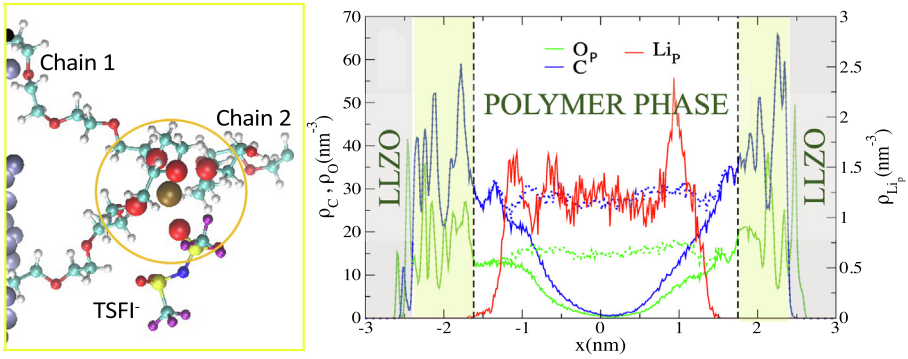
journal homepage: www.elsevier.com/locate/jcis

On the interfacial lithium dynamics in $Li_7La_3Zr_2O_{12}$:poly(ethylene oxide) (LiTFSI) composite polymer-ceramic solid electrolytes under strong polymer phase confinement

Mauricio Rincon Bonilla [a,*], Fabián A. García Daza [b], Henry A. Cortés [a], Javier Carrasco [c], Elena Akhmatskaya [a,d]

[a] BCAM - Basque Center for Applied Mathematics, Alameda de Mazarredo 14, E-48009 Bilbao, Spain
[b] Department of Chemical Engineering, The University of Manchester, Manchester M13 9PL, United Kingdom
[c] Centre for Cooperative Research on Alternative Energies (CIC energiGUNE), Basque Research and Technology Alliance (BRTA), Alava Technology Park, Albert Einstein 48, 01510 Vitoria-Gasteiz, Spain
[d] IKERBASQUE, Basque Foundation for Science, Plaza Euskadi 5, 48009 Bilbao, Spain

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 18 February 2022
Revised 30 April 2022
Accepted 11 May 2022
Available online 16 May 2022

Keywords:
Polymer-ceramic electrolytes
Interfacial lithium transport
Solid-state lithium ion batteries
Hybrid Monte Carlo
Umbrella sampling
Molecular dynamics

ABSTRACT

A better molecular-level understanding of Li^+ diffusion through ceramic/polymer interfaces is key to design high-performance composite solid-state electrolytes for all-solid-state batteries. By considering as a case study a composite electrolyte constituted by Li^+ conductive Ga^{3+} doped-$Li_7La_3Zr_2O_{12}$ (LLZO) garnet fillers embedded within a poly(ethylene oxide) and lithium bis(trifluoromethanesulfonyl) imide polymer matrix (PEO(LiTFSI)), we investigate Li^+ interfacial dynamics at conditions of high polymer confinement, with large filler particles in a fully amorphous polymer phase. Such confinement scenario is aimed to capture the conditions near the percolation threshold, at which conductivity enhancement is often reported. Using molecular dynamics simulations combined with the generalized shadow hybrid Monte Carlo method and umbrella sampling calculations, we explain why the hopping towards the polymer phase of the Li^+ sitting on the LLZO surface is thermodynamically hindered, while hopping of Li^+ from the polymer to the LLZO is kinetically slowed-down by rigidified polymer near the interface. In addition, we demonstrate how the overlap of LLZO-bound polymer chains at high confinement leads to a decrease of Li^+ diffusivity within the interstitial space. We put forward that these insights are relevant to interpret

* Corresponding author.
E-mail address: mrincon@bcamath.org (M.R. Bonilla).

https://doi.org/10.1016/j.jcis.2022.05.069
0021-9797/© 2022 Elsevier Inc. All rights reserved.

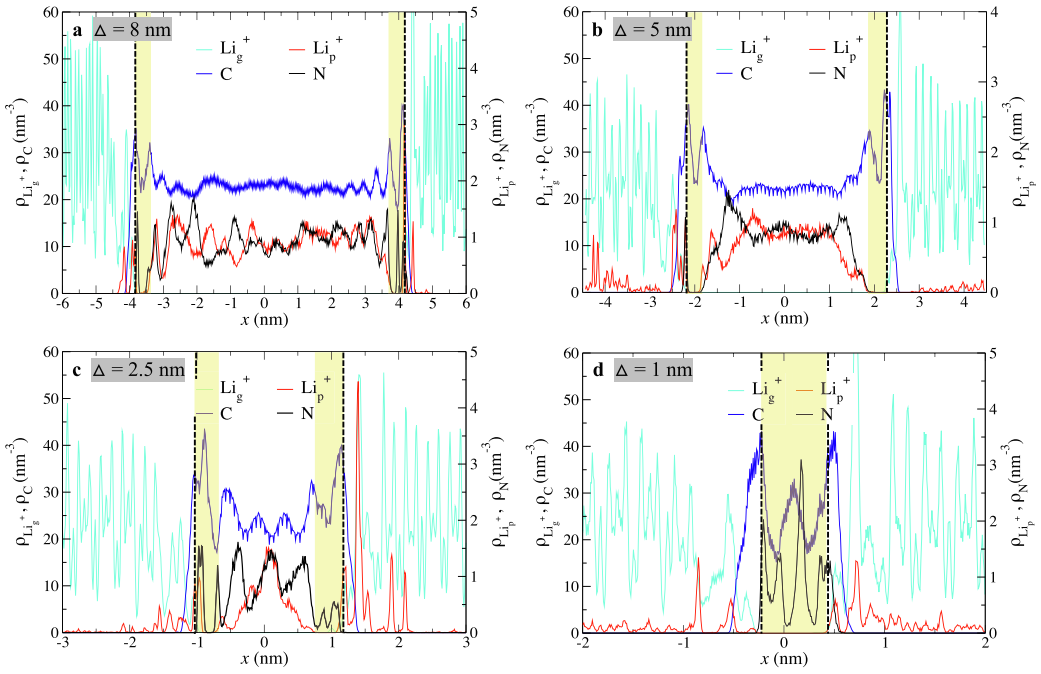


Fig. 2. Density profiles for selected atoms in the CSSE: C atoms from PEO (blue), N atoms from the $TFSI^-$ anions (black), Li^+ ions *originating* in the garnet phase, Li_g^+ (cyan), and the polymer phase, Li_p^+ (red). The curves were extracted at 450 K, for $N = 10$ and **a** $\Delta = 8$ nm, **b** $\Delta = 5$ nm, **c** $\Delta = 2.5$ nm, and **d** $\Delta = 1$ nm. The yellow bands represent Li^+-freelayers on the polymer side of the interface. The dashed lines indicate the limits of the garnet surfaces.

freedom available in the polymer phase, it is rather challenging to predict the average energy barrier associated with the transfer of a Li^+ ion from LLZO to PEO(LiTFSI) using only atomistic tools. However, by calculating the free energy profile, $F(x^*)$, for individual Li^+ ions along viable reaction coordinates, we can qualitatively elucidate the mechanistic origin of the density profiles, $\rho_{Li_p^+}$ and $\rho_{Li_g^+}$, depicted in Fig. 2. Here, coordinate x^* denotes distance from the garnet surface.

First, we randomly selected a Li^+ ion (denoted as Li*) from the garnet surface and run a steered MD run perpendicularly to the interface. Subsequently, umbrella sampling (US) calculations were performed by running constrained MD runs at selected windows along the steered MD trajectory (we chose a window every 0.05 nm). $F(x^*)$ was estimated through the Weighted Histogram Analysis Method (details and parameters for these calculations can be found in subSection 2.1).

Fig. 3a shows $F(x^*)$ for the extraction of a Li^+ ion from a randomly chosen surface site (designated site 1) at 450 K and for $\Delta = 8$ nm. The MD simulations at each window were run for time spans, t_w, of 10, 20 and 30 ns. Interestingly, the free energy curves tend to separate as x^* increases when $x^* > 0.35$ nm, but are nearly identical for $x^* < 0.35$ nm. The differences for $x^* > 0.35$ nm are indicative that $t_w = 30$ ns may be too short to accurately define $F(x^*)$ far from the interface where free polymer becomes available, while $t_w = 10$ ns is reliable for $x^* < 0.35$ nm. Fig. 3b shows $F(x^*)$ for the extraction of a Li^+ ion from two additional surface sites (sites 2 and 3) at the same conditions described before, and for $t_w = 10$ ns. While the shape of $F(x^*)$ differs for each site, the peak energy in all cases is between 0.7 and 0.8 eV, and is located at $x^* \sim 0.35$ nm. This indicates that the minimum work required for a Li^+ ion to traverse the Li^+ free layer is within this energy range, even though the curves themselves differ significantly as a consequence of the large variation in available atomistic and molecular environments.

The results above can be rationalized as follows. It is well understood that Li^+ ions are coordinated by a shell of ~ 5 polymer oxygens in amorphous PEO [29,30,51,52]. These oxygens are commonly associated with a single polymer chain [52]. Long-range diffusion occurs through Li^+ hopping between coordination shells in adjacent chains, and thus is highly correlated with the ability of the PEO chains to quickly fold and "surround" the diffusing ion [30].

For $x^* > 0.35$ nm, polymer segments are far enough from the garnet surface to freely fold and form a coordination shell around Li*. As x^* and t_w increase, the hopping likelihood within a given window increases, producing the divergence in $F(x^*)$ observed in Fig. 3a. Conversely, for $x^* < 0.35$ nm, the oxygen coordination shell is incomplete. To illustrate this, Figs. 3c and 3d depict the cumulative radial distribution function

$$n_O(r) = 4\pi\bar{\rho}_O \int_0^r r'^2 g_{O-Li*}(r')dr', \quad (1)$$

for the Li* from site 1 and both $O = O^{2-}$ (oxygens from the garnet) and $O = O_p$ (oxygens from the polymer phase). Here, $\bar{\rho}_O$ is the average density of O atoms and $g_{O-Li^*}(r)$ the radial distribution function for the $O - Li^*$ pair. At $x^* = 0$, Li* is coordinated in average by two O^{2-} atoms and one O_p atom within a 0.3 nm shell. Thus, Li* is undercoordinated with respect to both the polymer and the garnet phase. In spite of this, diffusion towards the polymer is strongly penalized, given the rapid increase in $F(0 < x^* < 0.35\text{nm})$ depicted in Fig. 3a. To explain this, we must recall that the coordination shell around a Li^+ atom is considerably tighter in the LLZO lattice than in bulk PEO. Indeed, Fig. 3c shows that the O^{2-} coordination shell has an average radius of 0.22 nm at $x^* = 0$, compared to the 0.3 nm shell formed by the O_p atoms (Fig. 3d). In addition, electrostatic effects generate vacancy-rich regions near the LLZO surface [8,19,23]. As a consequence, diffusion towards the garnet bulk is significantly more likely than towards the polymer phase.

From $x = 0$ to $x = 0.35$ nm, $n_{O^{2-}}|_{r=0.3\text{nm}}$ decreases to 0, while $n_{O_p}|_{r=0.3\text{nm}}$ increases to 4.8. The inset in Fig. 3c shows that the total oxygen coordination ($n_{Ot} = n_{O^{2-}}|_{r=0.3\text{nm}} + n_{O_p}|_{r=0.3\text{nm}}$) plateaus at $x^* = 0.35$ nm. Therefore, the Li^+ extraction barrier is the result of breaking the tight shell of garnet oxygens around Li* and transitioning to complete polymer coordination. At $x^* \sim 0.35$ nm, the

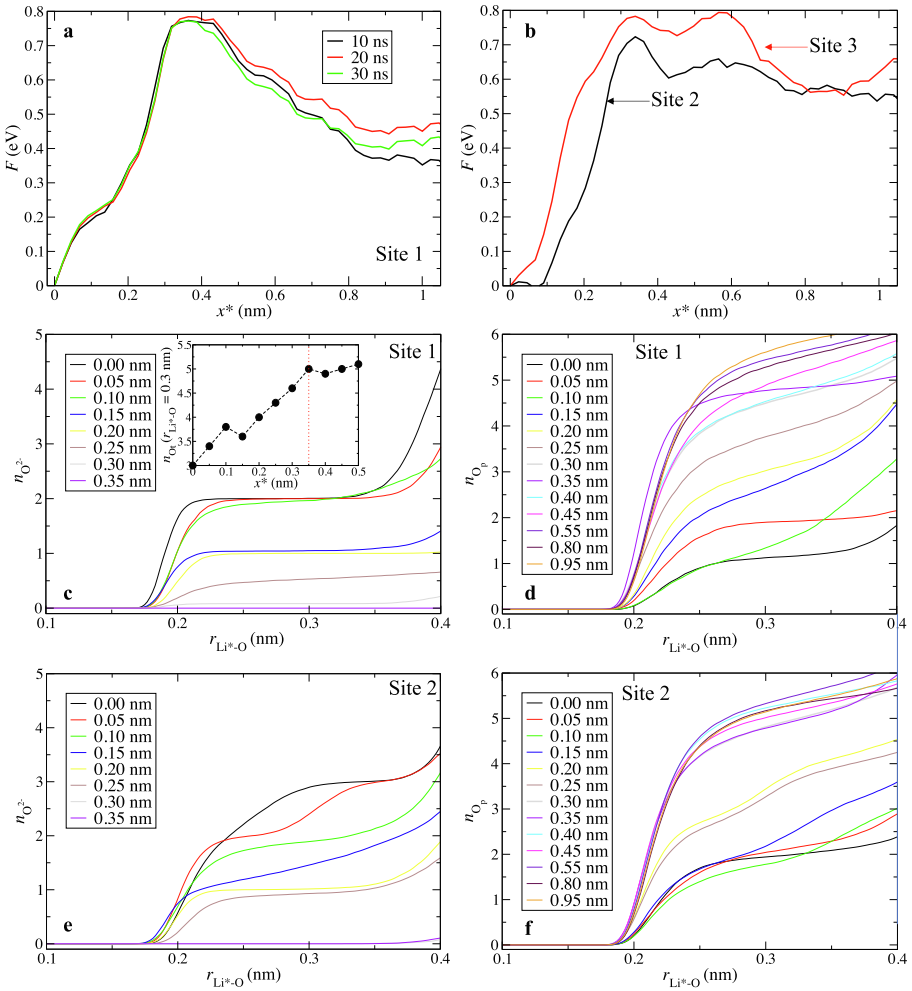


Fig. 3. **a** Free energy profile, $F(x^*)$, associated with the extraction of a selected Li^+ ion, Li*, from the garnet surface at 450 K and for $\Delta = 8$ nm, obtained through umbrella sampling (US) calculations. x^* denotes distance from the garnet surface. Windows in the US scheme were separated by 0.05 nm. The profiles for three simulation lengths per window ($t_w = 10$, 20, and 30 ns) are shown. **b** $F(x^*)$ associated with the extraction of selected Li^+ ions at two other randomly selected surface Li-sites ($t_w = 10$ ns). **c, d** Average number of garnet oxygens, $n_{O^{-2}}$, and polymer oxygens, n_{O_p}, respectively, within a shell of diameter r_{Li^*-O} around the Li^+ extracted from site 1, at several values of x^*. Inset: total coordination $n_{Ot} = n_{O^{2-}}|_{r_{Li^*-O}=0.3nm} + n_{O_p}|_{r_{Li^*-O}=0.3nm}$. **e, f** $n_{O^{-2}}$, and n_{O_p} within a shell of diameter r_{Li^*-O} around the Li^+ extracted from site 2 at several values of x^*.

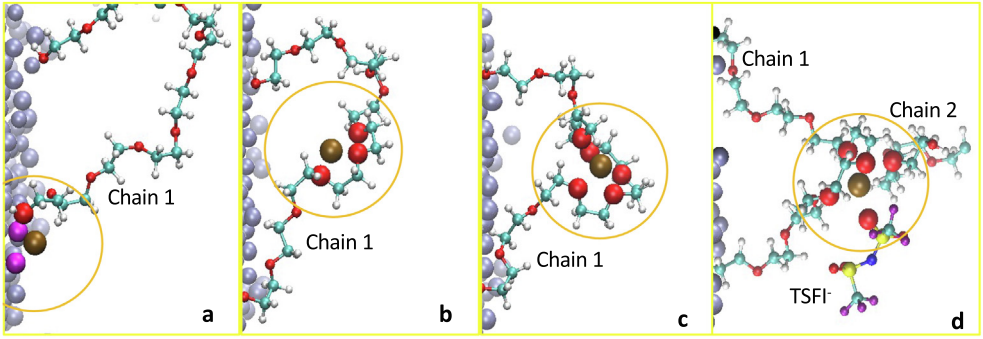


Fig. 4. Selected snapshots depicting the evolution of a Li^+ ion, Li* (brown sphere), extracted from the garnet. **a** At $x^* = 0$, Li* is (under) coordinated by 2 O^{2-} ions (magenta sphere) and one O_p atom (large red sphere). **b** At $x^* = 0.3$ nm, Li* is coordinated by 3 O_p atoms from a single bound chain (chain 1). **c** At $x^* = 0.38$ nm, Li* is fully coordinated by 5 O_p atoms from chain 1. **d** At $x^* = 0.5$ nm, Li* is simultaneously coordinated by chain 1, chain 2 (not bound) and a free $TSFI^-$ anion, reducing the entropic cost of Li* coordination.

electrostatic effect of the garnet oxygens is overcome by the interactions with O_p atoms. Nonetheless, it is costly for bound polymer chains to acquire configurations under which the Li* ion can be fully coordinated. For $x^* \gg 0.35$ nm, polymer chains and $TSFI^-$ ions that are not bound to the garnet can now form a coordination shell around Li*, lowering the entropic cost and reducing $F(x^*)$ (more significantly so for site 1 within the simulated t_w values). Fig. 4 schematically illustrates the evolution of the Li* extraction process.

The plots corresponding to $n_{O^{2-}}$ and n_{O_p} for the Li* from site 2 are shown in Figs. 3e and f. Clearly, the local environments of sites 1 and 2 differ. For instance, $n_{O^{2-}}|_{r=0.3nm} \sim 3$ nm for site 2 (comparing the curves for $x^* = 0$ and $x^* = 0.05$ nm in Figs. 3e reveals, however, that only 2 of the 3 O^{2-} ions are tightly bound to Li*; i.e., at an average distance of 0.22 nm). Even though these differences lead to substantial discrepancies in the shape of $F(x^*)$ for sites 1 and 2, similar results are obtained regarding the location and magnitude of the free energy peak. However, a key question remains unanswered: given that $F(x^* > 0.35\ nm) > F(0)$ up to $x^* = 1$ nm for all of the examined sites, why are there no more Li_p^+ ions transferring from the polymer to the garnet phase? In particular, $F(x^*)$ decreases very slightly within $0.35 < x^* < 1$ nm for Li* from sites 2 and 3, suggesting that such a transfer should occur quickly whenever these sites are available. Yet, the concentration of Li_p^+ in the garnet is insignificant for $\Delta = 8$ nm (see Fig. 2a). The reason is not thermodynamic, but kinetic, and is described in the following section.

5. Effect of rigidified polymer layer

The formation of a rigidified layer of polymer at the interface, due to attractive interactions between the polymer and filler promoted by stronger polymer-filler interaction compared to polymer-polymer interaction is a well-known problem in mixed matrix membranes for molecular sieving [49,53,54]. While its presence was reported in CSSEs incorporating passive fillers and amorphous PEO [27,52], only recently was it theoretically investigated on CSSEs involving active fillers [29].

The impact of the rigidified layer on interfacial transport can be further understood through the equilibrium density profiles. However, we found that considering the entire thickness of simulation box was insufficiently clear (see Section S5 in the SI for further details). Thus, we examined the density profiles within two slabs of thickness $\Delta y = 1$ nm centered at $y_1 = 1$ nm and $y_2 = 3$ nm, as schematically shown in Fig. 5. The resulting density profiles are shown in Fig. 6. The solid lines represent the densities of C (blue) and O_p (green) atoms from bound PEO chains only, while the dotted lines represent the densities of C and O_p for both free and bound chains. The regions where these lines coincide correspond to the rigidified polymer layers: in this region, all C/O_p atoms belong to bound chains, i.e., there are no segments of free chains available. We have highlighted the rigidified layers in yellow. In practice, they were identified as the areas in which the dashed and solid lines differ by less than 1 %.

Figs. 6a-d show that, for $\Delta = 5$ nm, the Li_p^+ density profiles (red line) vanish within the yellow regions. Thus, the rigidified layer matches almost exactly the Li^+ free layer. The present result shows that free PEO chains are significantly better at stabilizing Li_p^+ ions than bound PEO. The reason behind it is associated with the higher flexibility of the free chains: beyond the rigidified layer, it is relatively easy for one (or more) free chain(s) in the vicinity of a Li_p^+ ion to quickly and effectively rearrange to form a stabilizing oxygen coordination shell. We also note that the overlap between polymer shells observed in Figs. 6c,d ($\Delta = 5$ nm, $N = 20$) is much more significant than in Figs. 6a,b ($\Delta = 5$ nm, $N = 10$). This results in a greater heterogeneity of polymer mobilities within the central region in Figs. 6c,d, leading in turn to greater heterogeneity in $\rho_{Li_p^+}$.

The thickness of the Li^+ free layer (defined by the thickness of the yellow bands) varies between 0.5 and 0.7 nm in Figs. 6a and b, and 0.4 and 0.6 nm in Figs. 6c and d. Nonetheless, its boundary is far from $x^* = 0.35$ nm, where the peak in $F(x^*)$ is located. That is, in order to make use of the downhill path in the free energy profile to reach the garnet surface, the Li_p^+ must first penetrate the rigidified layer. We can qualitatively visualize this by considering the scenario in which the Li_p^+ ion in Fig. 4d evolves towards the configuration shown in Fig. 4c: because Chain 2 is significantly more flexible than Chain 1, it can quickly "shuffle" around to capture Li_p^+ within a coordination shell, never giving Chain 1 enough time to form a "receiving" coordination shell to which Li_p^+ can hop into. Therefore, the rate at which Li_p^+ ions are adsorbed by LLZO is significantly slowed down due to the poor kinetics of the bound polymer, despite the considerable thermodynamic benefit. As a matter of fact, a large fraction of the Li_p^+ ions that were adsorbed by the garnet in our simulations were located, at $t = 0$, within the rigidified layer. We note that these conclusions apply for $\Delta \geqslant 5$ nm and with $N \leq 20$, when free polymer is available. However, we can assume that for smaller Δ and/or sufficiently large N, no free chains will remain. This situation is akin to that shown in Figs. 6e and f.

For $\Delta = 2.5$ nm and $N = 20$, all polymer chains are bound to the garnet (Figs. 6e,f). In the absence of free chains, any Li_p^+ remaining in the polymer phase must be confined in the central region, at least 0.35 nm away from either surface (as deduced from Figs. 3a,b). At an equal distance from each surface one would expect the polymer to be most mobile, that explains the central peak in $\rho_{Li_p^+}$. In fact, we have estimated that each Li_p^+ in the polymer phase is coordinated by an average of two different PEO chains at any given time. Thus, the peak in $\rho_{Li_p^+}$ near $x = 0$ corresponds to lithium ions that are "trapped" in coordination shells from polymer chains bound to opposite garnet surfaces.

Figs. 6e,f also reveal significant adsorption peaks in $\rho_{Li_p^+}$, 3–5 times more intense than those in Figs. 6a - d. Moreover, the Li^+ free layer is 0.8–1 nm thick. This can be explained by noting that, under such degree of confinement, it will be significantly easier for Li_p^+ ions to find the downhill path in the free energy landscape towards the garnet surface. Indeed, in the most extreme confinement illustrated in Fig. 2d for $\Delta = 1.0$ nm, not a single Li_p^+ is left in the polymer phase. We note that $\Delta = 2.5$ nm and $\Delta = 1.0$ nm correspond to large volume fractions, which are associated with low conductivity and brittleness [17]. However, we will use these scenarios of

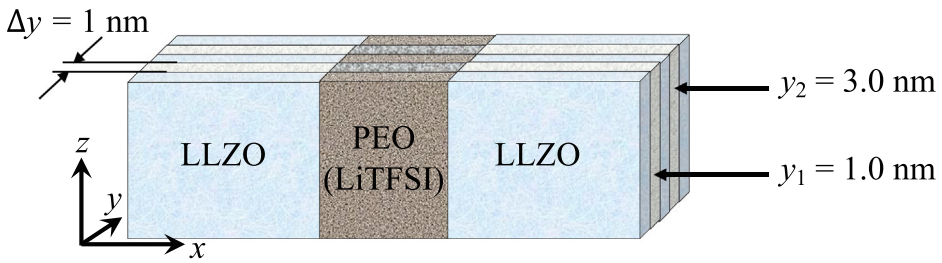


Fig. 5. Density profiles $\rho(x)$ in Fig. 6 are measured within slabs of thickness $\Delta y = 1$ nm centered at $y_1 = 1$ nm and $y_2 = 3$ nm.

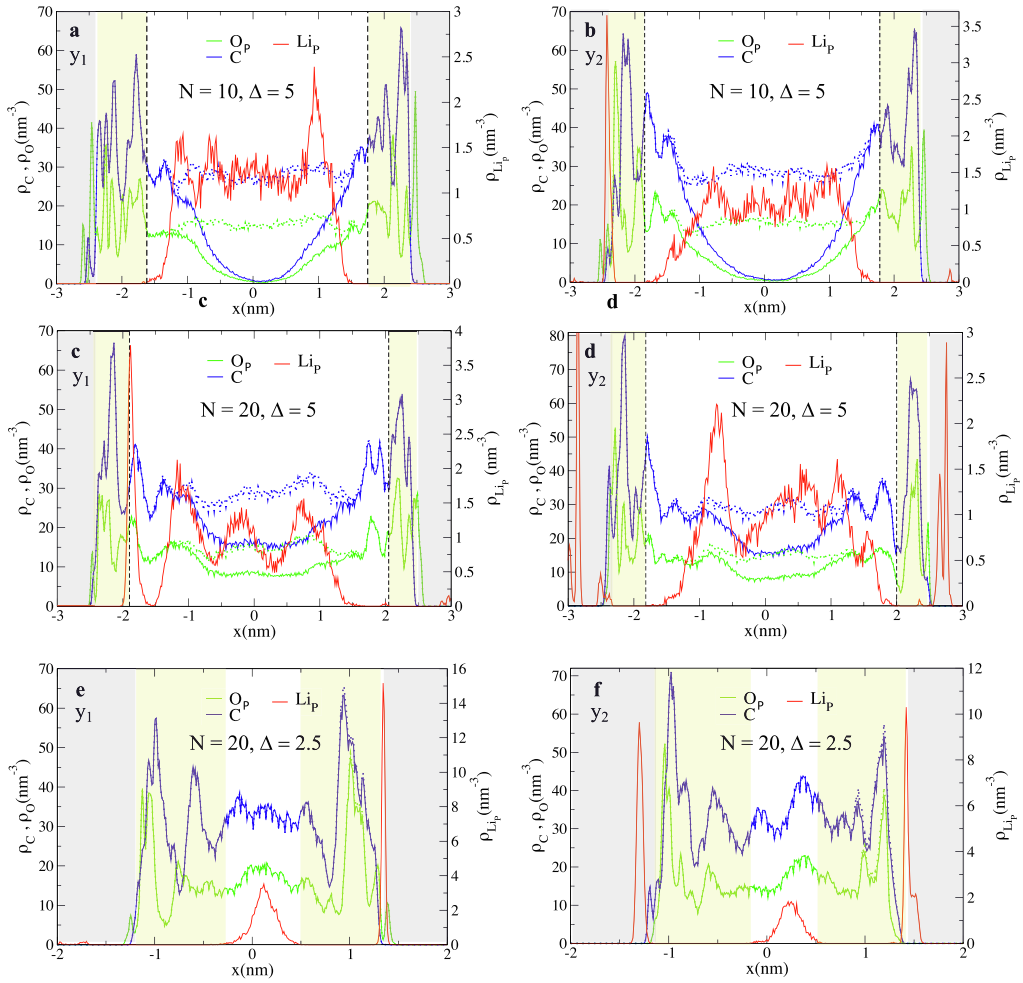


Fig. 6. Density profiles $\rho_X(x)$ for $X = \mathrm{Li}_p^+$, C and O_p and within slabs of thickness 1 nm in the y direction, centered at $y_1 = 1$ nm and $y_2 = 3$ nm (see Fig. 5). The solid blue (green) line represents C (O_p) atoms from bound PEO chains, while the dotted blue (green) line represents C (O_p) atoms from all chains. The dashed vertical lines designate the lowest and highest x values at which corresponding solid and dotted lines merge. The yellow bands indicate the rigidified layer (where all available C and O_p belongs to a bound chain). The gray areas indicate the location of the garnet. The following configurations were examined: **a** $N = 10, \Delta = 5$ nm (centered at y_1). **b** $N = 10, \Delta = 5$ nm (centered at y_2). **c** $N = 20, \Delta = 5$ nm (centered at y_1). **d** $N = 20, \Delta = 5$ nm (centered at y_2). **e** $N = 20, \Delta = 2.5$ nm (centered at y_1). **f** $N = 20, \Delta = 2.5$ nm (centered at y_2).

extreme confinement to show what happens to Li^+ diffusion when bound polymer shells around filler particles are close enough to overlap.

6. Li^+ diffusion under confinement

In this section, we investigate Li^+ diffusion in the polymer phase under strong polymer confinement ($\Delta \sim l_p$ and $l_p/d_p \ll 1$, where l_p is the mean end-to-end distance of the PEO chains and d_p the particle diameter). Li^+ diffusion in the x direction is anomalous, due to the boundaries imposed by the garnet and the rigidified polymer layer. Thus, we focus on the diffusivity, D, in the $y-z$ plane, following the formulation proposed by Liu et al. [55] for lateral diffusion near surfaces (a brief summary can be found in Section S6 of the SI). To avoid the Li^+ free layer, we measure D within a region Δ^* at least 0.8 nm from the garnet surface (Fig. 7a).

Fig. 7b depicts the mean square displacement in the $y-z$ plane, $\mathrm{MSD}_{y,z}(t)$, for systems having $N = 20$, and $\Delta = 5$ and 8 nm at 450 K. The slope of the curves reaches unity at ~ 4 ns, indicating Fickian diffusion. Therefore, D can be estimated for this study through Einstein's diffusion equation using the appropriate modifications to account for finite residence times [55]. The $\mathrm{MSD}_{y,z}(t)$ curves for all other investigated systems follow approximately the same behavior, although for $\Delta = 2.5$ nm Fickian diffusion is only registered after ~ 20 ns. There are clear differences in $\mathrm{MSD}_{y,z}(t)$ between $\Delta = 8$ nm and $\Delta = 5$ nm, despite the polymer length being kept constant. Below, we explain how such differences originate from the overlapping of polymer shells bound to opposing surfaces.

Previous experiments and simulations studies [30,56] showed that the Li^+ diffusivity decreases with N by a factor of 2–3 for $20 < N < 100$ in bulk PEO(LiTFSI), and plateaus thereafter. Interestingly, Fig. 8 reveals that although the ratio $r_D = D(N = 10)/D(N = 20)$ is always above unity (as expected), its magnitude is non monotonically dependent on Δ. While $r_D = 1.1$ for $\Delta = 2.5$ nm and 1.3 for $\Delta = 8$ nm, it raises to 4.5 for $\Delta = 5$ nm. To explain this behavior, let us recall that the "polymer shell" of a given surface corresponds to the layer with minimum average thickness containing all the polymer chains bound to that surface. The rigidified layer, comprising the region in which "all" available polymer is physically bound to the garnet is part of the polymer shell. In fact, the polymer shell can be significantly thicker than the rigidified layer (the thickness of the polymer shell is $\sim N^{3/5}$ according to Lee et al. [57]).

Fig. 9 illustrates the number density profiles for oxygen (O_p) and carbon (C) atoms from PEO chains bound to the garnet surfaces, for $\Delta = 2.5, 5.0$, and 8.0 nm and for $N = 10$ and 20. We separate the profiles associated with the left and right interfaces, allowing us

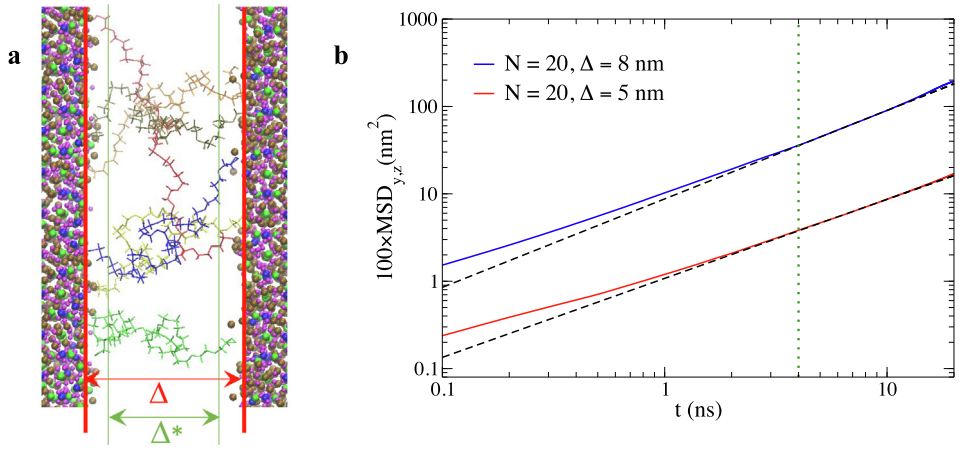


Fig. 7. **a** When large garnet surfaces are sufficiently close ($\Delta \sim l_p$ and $l_p/d_p \ll 1$), individual polymer chains may interact with both interfaces as shown in the figure (free polymer chains or those bound to only one interface are not shown for clarity). The diffusivities are estimated within a region Δ^* at least 0.8 nm away from the garnet surfaces, with the goal of avoiding the rigidified polymer layers and the adsorption field from the LLZO. **b** log-log plot of the mean square displacement for Li_p^+ ions within Δ^*, at 450 K, and for systems with $N = 20$ and both $\Delta = 5$ and 8 nm. Past the vertical dotted line, the slope of the curves is approximately one.

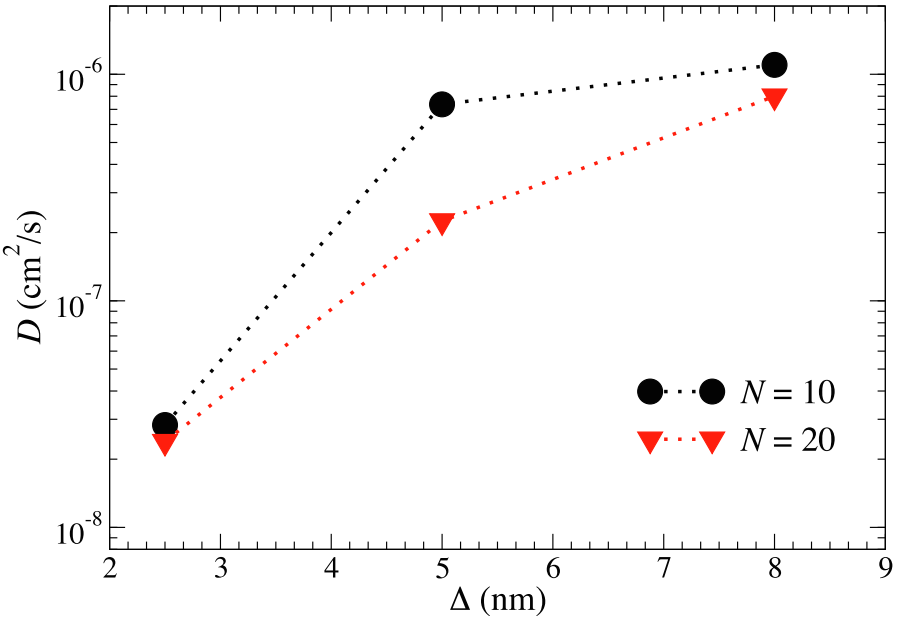


Fig. 8. Diffusion coefficient, D, as a function of Δ, at 450 K and for PEO chains with $N = 10$ and $N = 20$.

to visualize the thickness of the polymer shells for each garnet surface. Figs. 9a,b indicate that for $\Delta = 2.5$ nm, the polymer shells from each surface largely overlap for both $N = 10$ and 20. In either case, D is about two orders of magnitude below the bulk diffusivity of Li^+ in bulk PEO(LiTFSI) ($D_{Bulk} = 3.3 \times 10^{-6}$ cm^2/s (2.1×10^{-6} cm^2/s) for $N = 10$ (20) at 450 K) [29]. Conversely, Figs. 9e,f reveal that for $\Delta = 8.0$ nm, the polymer shells from each surface do not interact, and $D \sim 0.3D_{Bulk}$ regardless of N. Yet, the r_D values for $\Delta = 8.0$ nm and $\Delta = 2.5$ nm remain very close. Moreover, for $\Delta = 5.0$ nm, there is polymer shell overlapping when $N = 20$ (Figs. 9d) and no overlapping when $N = 10$ (Figs. 9c), which produces a large r_D of 4.5. In previous work [29], we hypothesized that overlapping of the polymer shells from adjacent particles was responsible for the pronounced drop in Li^+ conductivity experimentally observed at high volume fractions ($\omega \sim 45\%$ for particles 1 μm in diameter and 10^6 g/mol PEO) [17]. While mechanical measurements and particle packing arguments supported this hypothesis, no atomistic evidence was provided at the time. The present results constitute definitive atomistic confirmation: overlapping of polymer shells leads to a dramatic reduction in the Li^+ diffusivity (by a factor of 4.5 in our case). As a consequence, a percolation-like drop in the ionic conductivity should occur when the "filler + polymer shell" pseudo-particles overcome random close packing [29].

The segmental mobility of the polymer chains can be characterized by the autocorrelation function for the C–O–C–C dihedrals, $C_{\phi\phi}$: [58,27]

$$C_{\phi\phi}(t) = \frac{\langle cos\phi(t)cos\phi(0)\rangle - \langle cos\phi(0)\rangle^2}{\langle cos\phi(0)cos\phi(0)\rangle - \langle cos\phi(0)\rangle^2}. \quad (2)$$

Fig. 10 depicts the decay of $C_{\phi\phi}$ for bound PEO chains at three combinations of Δ and N. The slowest decay (and hence the lowest segmental mobility) corresponds to $\{\Delta = 2.5$ nm, $N = 10\}$. This is expected, as a small fraction of these chains is able to reach the opposite wall (see Fig. 9a). However, there is little difference between $C_{\phi\phi}$ for $\{\Delta = 5$ nm, $N = 20\}$, and $\{\Delta = 5$ nm, $N = 10\}$. This indicates that polymer shell overlapping does not reduce the Li^+ diffusivity by decreasing the mobility of the overlapping bound

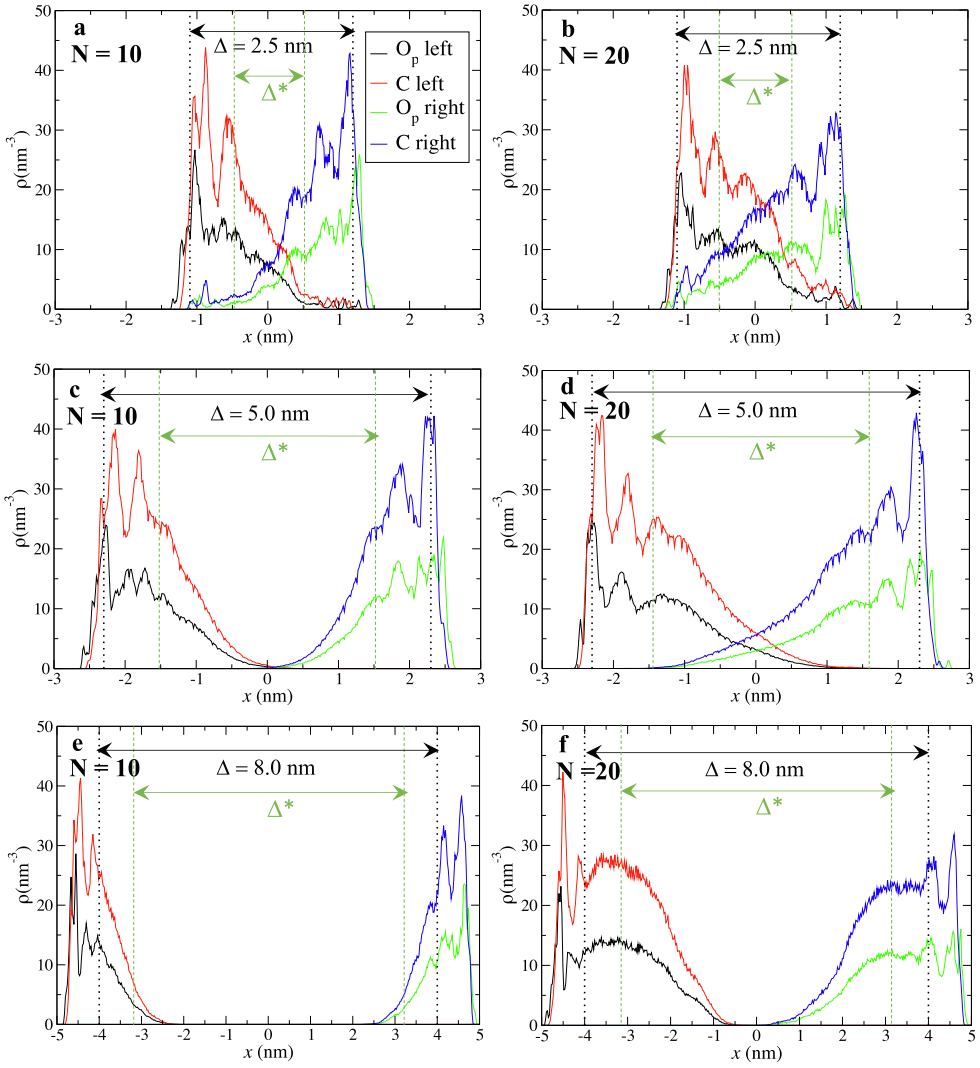


Fig. 9. Density profiles of oxygen (O_p) and carbon (C) atoms from PEO chains bound to the garnet surface, at 450 K and for $N = 10$ and 20 at $\Delta = 2.5, 5.0, 8.0$ nm (legend for all panels is available in panel **a**). Δ^* represents the region within which the values of D in Fig. 8 were calculated. We disaggregate the profiles corresponding to chains bound to the left and right surfaces. The polymer shells from each side interpenetrate in **a**, **b** and **d**, impacting Li^+ diffusion.

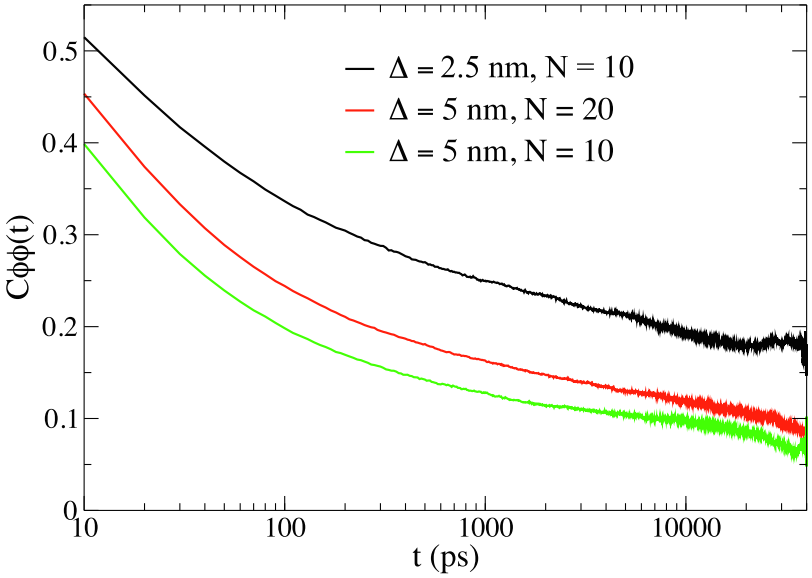


Fig. 10. Variation of $C_{\phi\phi}$ with time for bound polymer chains at three different combinations of Δ and N.

chains. Simply, there is less free polymer available in the overlap region at the given density, lowering the frequency of interchain jumps.

7. Impact of confinement on ionic conduction

Several authors have reported SCEs capable of enhancing the ionic conductivity σ (which is roughly proportional to D [29]) in CSSEs when the filler particles are sufficiently close [23,60]. We do not find evidence for such effects in our simulations. However, SCEs appear to manifest more strongly in CSSEs with small nanofillers ($\sim$ 10 nm), for which the conditions set out in this study ($\Delta \sim l_p$ and $l_p/d_p \ll 1$) are not met.

Conductivity enhancement below the PEO melting temperature $T_m (\sim 68\ ^\circ C)$ may also be caused by loss of polymer crystallinity around the filler particles [61]. Because amorphous PEO is substantially more conductive than crystalline PEO, the formation of a spanning cluster of amorphous polymer leads to an increase in σ with ω for $T < T_m$, up to a critical value ω_c. This behavior has traditionally been modeled through effective medium theory (EMT) - based models [24,50,62]. EMT assumes the particles comprise an insulating core and a highly conductive shell, and predict reasonably well the variation of σ with ω in several CSSEs with passive fillers [24,62]. Nonetheless, they also predict a dramatic decrease in conductivity for $\omega > \omega_c$, which is not observed in LLZO:PEO (LiTFSI). For instance, Zheng et al. [59] showed that σ decreases slowly for $\omega > \omega_c$ in LLZO:PEO(LiTFSI) CSSEs with 40 and 400 nm particles at 30 °C, in contrast with typical EMT expectations (Fig. 11a).

The results presented so far may shed some light into this behavior. We present in Fig. 11b an alternative particle model, in which the filler particles are surrounded by a layer of amorphous polymer. The polymer shell has a conductivity σ_b and is contained within this amorphous layer, leaving a second outer layer of free amorphous polymer with conductivity σ_f. Assuming interfacial Li^+ exchange contributes negligibly to ionic conduction due to the high surface resistance [18,19] and that $\sigma_f \gg \sigma_b \gg \sigma_c$ [29,59], where σ_c is the conductivity of crystalline PEO, we can explain the slow decay in the conductivity (σ) of the CSSE as follows: for $\omega < \omega_c$, the layers of free amorphous polymer begin to overlap as ω increases, leading to overall conductivity enhancement in the CSSE. At $\omega = \omega_c$, a spanning cluster of free amorphous polymer forms that produces a maximum in σ. However, such a maximum is tempered by the overlapping of polymer shells, which was shown to negatively impact Li^+ diffusion. Increasing ω enhances the fraction of overlapping polymer shells, further reducing the conductivity while still allowing a relatively high ionic flow as compared to that in the crystalline polymer.

It has been found that at $\omega \sim 33\%$ a spanning cluster of LLZO particles may also arise, allowing Li^+ transport through the garnet phase [18]. This latter diffusion path is known to be hindered by high grain boundary resistances, but would still contribute to dampening the decline in the conductivity. Thus, the significant drop in $\sigma(T < T_m)$ predicted by several EMT models past ω_c may not occur in CSSEs with active fillers that can develop a bound polymer shell. Additional quantitative work will be required to validate our double layer model, but it is a promising framework to uncover this intriguing phenomenon.

8. Conclusions

This work investigates the Li^+ spacial distribution and dynamics in LLZO:PEO(LiTFSI) CSSEs at conditions of high polymer confinement. Specifically, we focus on high temperature systems (450 K) and large filler particles ($\Delta \sim l_p$ and $l_p/d_p \ll 1$), in which the polymer phase is fully amorphous and the particles can be assumed to be semi infinite perpendicularly to the interface. By employing umbrella sampling calculations, we show that the free energy profile associated with the extraction of a Li^+ ion from the garnet surface is highly heterogeneous and may require very long MD runs at each window in the umbrella sampling scheme. However, the maximum free energy is surprisingly uniform at 0.7–0.8 eV for the three studied cases, and occurs $\sim$ 0.35 nm from the garnet surface. Moreover, the free energy profile is non symmetric, suggesting a thermodynamic drive towards Li^+ adsorption into the garnet phase. We show that surface Li^+s are coordinated by a combination of polymer and garnet oxygens, with the latter forming a tighter shell. It is only beyond 0.35 nm from the garnet surface that the polymer coordination shell reaches its typical bulk density of $\sim$ 5 oxygens per Li^+, leading to a slight decrease in the free energy. We show that the thermodynamic trend towards Li^+ uptake by the garnet is countered by a kinetic effect: free polymer chains are significantly more mobile than garnet - bound chains. Hence, they can quickly adapt to coordinate and stabilize incoming Li^+ ions. As a consequence, Li^+ jumps from a free chain to an adjacent bound chain are far less frequent than those to adjacent free chains, minimizing the actual amount of Li^+ reaching the LLZO surface. Only at very high garnet contents we predict Li^+ adsorption to be thermodynamically controlled, but such contents are, in general, of no practical interest.

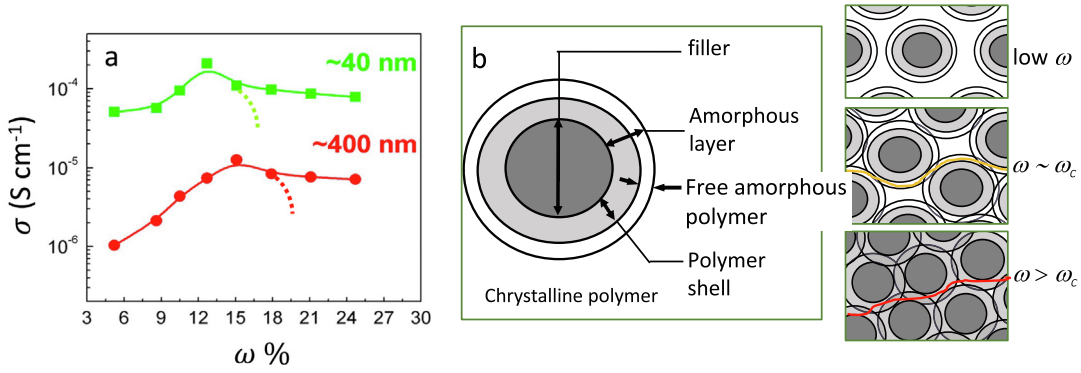


Fig. 11. **a** The conductivity as a function of Ta-doped LLZO content for 40 and 400 nm particles in LLZO:PEO(LiTFSI) CSSEs at 30 °C. The solid line interpolates the experimental data, while the dotted line shows the typical curve shape obtained from core-shell EMT models. Adapted from Zheng et al. [59], Copyright Elsevier (2016). **b** Left panel: schematic of the proposed percolation process at $T < T_m$. The filler particles are surrounded by a layer of highly conducting amorphous polymer. The polymer shell (i.e., the collection of polymer chains that are physically bound to the filler particle) is contained within the amorphous layer. Right-top panel: at low ω, conduction through the crystalline phase dominates. middle-panel: at ω_c, a spanning cluster of overlapping amorphous layers arises, with minimal overlap between polymer shells. Bottom-panel: for $\omega > \omega_c$, polymer shells begin to overlap, reducing the conductivity at a relatively slow pace as ω increases.

We also demonstrate that the overlapping polymer shells lead to a decrease in Li^+ diffusivity within the interstitial space. This was originally hypothesized in our previous work [29] but remained untested. In addition, we fully discard conductivity-enhancing SCEs at the specific conditions studied here. They may still be present in CSSEs incorporating small nanoparticles, at lower temperatures or with methyl-terminated PEO chains. Moreover, additional conditions may be required at the interface for these effects to manifest (such as residual Li_2ClO_3 impurities, which is largely present in typical synthesis approaches [18,20]).

Understanding Li^+ diffusion in LLZO:PEO(LiTFSI) CSSEs above the polymer melting point, T_m, and at conditions of high polymer confinement may still provide valuable insights to interpret the variation of ionic conductivity, σ, with volume fraction, ω, below T_m. This is because filler particles are thought to be surrounded by a layer of amorphous polymer even at room temperature, and consequently the current model may be adequate at sufficiently high levels of confinement. Based on this premise, we qualitatively describe the variation of σ with ω reported in Zheng et al. [59]. In summary, we propose that the amorphous layer can be subdivided into two regions: bound and free amorphous polymer. While this is an oversimplification, it captures the fact that the closer a PEO chain is to the garnet the least flexible it is. This model qualitatively explains the relatively slow decay in ionic conductivity past the critical percolation volume fraction ω_c, avoiding the power-law like decay obtained from the traditional core-shell EMT models that have been employed to explain conductivity enhancement in CSSEs with passive fillers. Yet, further quantitative work is required to validate this hypothesis.

While in any given system both garnet surfaces differ in the distribution of ions, key features such as the thickness of the Li^+-free layer (which largely coincides with the thickness of the rigidified polymer layer), differ by less than 1 nm between the opposing sides. Moreover, the shapes and thicknesses of the polymer shells bound to either surface are largely symmetric. Because these are the two most significant features impacting the Li^+ diffusivity under conditions of high confinement (at a constant LiTFSI concentration), we expect that the differences between the garnet surfaces do not alter the main conclusions from this study. Still, we recognize that local differences in the garnet surface must be impactful. A systematic examination of this effect should be investigated in future work.

Extensive simulations by Brooks et al. [30] have shown that $N = 100$ suffices to accurately predict the Li^+ diffusivity in bulk PEO(LiFTSI) for system with high molecular weight polymer. However, this may not be the case in the presence of inorganic inclusions. In that regard, our results constitute a qualitative exploration of the phenomena occurring in the vicinity of the filler surface. Examining large, $N \geqslant 100$ polymer chains should be explored in future works by using, for example, coarse-grained MD methods.

Upcoming studies will also focus on studying CSSEs below T_m, employing, for example, kinetic Monte Carlo methods to predict Li^+ diffusion through crystalline PEO.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We acknowledge the financial support by the Ministerio de Economía y Competitividad (MICINN) of the Spanish Government through BCAM Severo Ochoa accreditation SEV-2017-0718 and PID2019-104927 GB-C22 grant. This work was supported by the BERC 2022-2025 Program and by ELKARTEK Programme, grants KK-2021/00064 and KK-2021/00022, funded by the Basque Government. The research is partially supported by the Basque Government under the grant Ikerbasque COVID-19. This work has been possible thanks to the computing infrastructure of the i2BASQUE academic network, DIPC Computer Center and the technical and human support provided by IZO- SGI SGIker of UPV/EHU.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jcis.2022.05.069.

References

[1] J. Janek, W.G. Zeier, A solid future for battery development, Nat. Energy 1 (9) (2016) 16141, https://doi.org/10.1038/nenergy.2016.141.
[2] Q. Zhao, S. Stalin, C.Z. Zhao, L.A. Archer, Designing solid-state electrolytes for safe, energy-dense batteries, Nat. Rev. Mater. 5 (3) (2020) 229–252, https://doi.org/10.1038/s41578-019-0165-5.
[3] T. Placke, R. Kloepsch, S. Dühnen, M. Winter, Lithium ion, lithium metal, and alternative rechargeable battery technologies: the odyssey for high energy density, J. Solid State Electrochem. 21 (7) (2017) 1939–1964, https://doi.org/10.1007/s10008-017-3610-7.
[4] P. Albertus, V. Anandan, C. Ban, N. Balsara, I. Belharouak, J. Buettner-Garrett, Z. Chen, C. Daniel, M. Doeff, N.J. Dudney, B. Dunn, S.J. Harris, S. Herle, E. Herbert, S. Kalnaus, J.A. Libera, D. Lu, S. Martin, B.D. McCloskey, M.T. McDowell, Y.S. Meng, J. Nanda, J. Sakamoto, E.C. Self, S. Tepavcevic, E. Wachsman, C. Wang, A.S. Westover, J. Xiao, T. Yersak, Challenges for and pathways toward Li-metal-based all-solid-state batteries, ACS Energy Lett. 6 (4) (2021) 1399–1404, https://doi.org/10.1021/acsenergylett.1c00445.
[5] S. Sen, E. Trevisanello, E. Niemöller, B.X. Shi, F.J. Simon, F.H. Richter, The role of polymers in lithium solid-state batteries with inorganic solid electrolytes, J. Mater. Chem. A 9 (2021) 18701–18732, https://doi.org/10.1039/D1TA02796D.
[6] T. Famprikis, P. Canepa, J.A. Dawson, M.S. Islam, C. Masquelier, Fundamentals of inorganic solid-state electrolytes for batteries, Nat. Mater. 18 (12) (2019) 1278–1291, https://doi.org/10.1038/s41563-019-0431-3.
[7] V. Thangadurai, S. Narayanan, D. Pinzaru, Garnet-type solid-state fast Li ion conductors for Li batteries: Critical review, Chem. Soc. Rev. 43 (2014) 4714–4727, https://doi.org/10.1039/C4CS00020J.
[8] M.R. Bonilla, F.A. García Daza, M. Fernández-Pendás, J. Carrasco, E. Akhmatskaya, Multiscale modelling and simulation of advanced battery materials, in: M. Cruz, C. Parés, P. Quintela (Eds.), Progress in Industrial Mathematics: Success Stories, Springer International Publishing, Cham, 2021, pp. 69–113.
[9] F. Aguesse, W. Manalastas, L. Buannic, J.M. Lopez del Amo, G. Singh, A. Llordés, J. Kilner, Investigating the dendritic growth during full cell cycling of garnet electrolyte in direct contact with Li metal, ACS Applied Materials & Interfaces 9 (4) (2017) 3808–3816, https://doi.org/10.1021/acsami.6b13925.
[10] K. Hofstetter, A.J. Samson, S. Narayanan, V. Thangadurai, Present understanding of the stability of Li-stuffed garnets with moisture, carbon dioxide, and metallic lithium, J. Power Sources 390 (2018) 297–312, https://doi.org/10.1016/j.jpowsour.2018.04.016.
[11] M. Vervaeke, G. Calabrese, Prospective design in the automotive sector and the trajectory of the bluecar project: An electric car sharing system, Int. J. Veh. Des. 68 (2015) 245, https://doi.org/10.1504/IJVD.2015.071083.
[12] T. Yang, J. Zheng, Q. Cheng, Y.Y. Hu, C.K. Chan, Composite polymer electrolytes with $Li_7La_3Zr_2O_{12}$ garnet-type nanowires as ceramic fillers: Mechanism of conductivity enhancement and role of doping and morphology, ACS Appl. Mater. Interfaces 9 (26) (2017) 21773–21780, https://doi.org/10.1021/acsami.7b03806.
[13] H. Xu, P.H. Chien, J. Shi, Y. Li, N. Wu, Y. Liu, Y.Y. Hu, J.B. Goodenough, High-performance all-solid-state batteries enabled by salt bonding to perovskite in poly(ethylene oxide), Proc. Natl. Acad. Sci. 116 (38) (2019) 18815–18821, https://doi.org/10.1073/pnas.1907507116.
[14] X. Yu, A. Manthiram, A review of composite polymer-ceramic electrolytes for lithium batteries, Energy Storage Mater. 34 (2021) 282–300, https://doi.org/10.1016/j.ensm.2020.10.006.
[15] F. Langer, I. Bardenhagen, J. Glenneberg, R. Kun, Microstructure and temperature dependent lithium ion transport of ceramic-polymer composite electrolyte for solid-state lithium ion batteries based on garnet-type $Li_7La_3Zr_2O_{12}$, Solid State Ionics 291 (2016) 8–13, https://doi.org/10.1016/j.ssi.2016.04.014.
[16] M. Keller, G.B. Appetecchi, G.T. Kim, V. Sharova, M. Schneider, J. Schuhmacher, A. Roters, S. Passerini, Electrochemical performance of a solvent-free hybrid ceramic-polymer electrolyte based on $Li_7La_3Zr_2O_{12}$ in P(EO)-LiTFSI, J. Power Sources 353 (2017) 287–297, https://doi.org/10.1016/j.jpowsour.2017.04.014.
[17] J. Zagórski, J.M. López del Amo, M.J. Cordill, F. Aguesse, L. Buannic, A. Llordés, Garnet-polymer composite electrolytes: New insights on local Li-ion dynamics

and electrodeposition stability with Li metal anodes, ACS Appl. Energy Mater. 2 (3) (2019) 1734–1746, https://doi.org/10.1021/acsaem.8b01850.
[18] J. Zheng, Y.Y. Hu, New insights into the compositional dependence of Li-ion transport in polymer-ceramic composite electrolytes, ACS Appl. Mater. Interfaces 10 (4) (2018) 4113–4120, https://doi.org/10.1021/acsami.7b17301.
[19] D. Brogioli, F. Langer, R. Kun, F. La Mantia, Space-charge effects at the $Li_7La_3Zr_2O_{12}$/poly(ethylene oxide) interface, ACS Appl. Mater. Interfaces 11 (12) (2019) 11999–12007, https://doi.org/10.1021/acsami.8b19237.
[20] A. Gupta, J. Sakamoto, Controlling ionic transport through the PEO-LiTFSI/LLZTO interface 28 (2) (2019) 63–69, https://doi.org/10.1149/2.F06192if.
[21] J. Zhang, N. Zhao, M. Zhang, Y. Li, P.K. Chu, X. Guo, Z. Di, X. Wang, H. Li, Flexible and ion-conducting membrane electrolytes for solid-state lithium batteries: Dispersion of garnet nanoparticles in insulating polyethylene oxide, Nano Energy 28 (2016) 447–454, https://doi.org/10.1016/j.nanoen.2016.09.002.
[22] W. Liu, S.W. Lee, D. Lin, F. Shi, S. Wang, A.D. Sendek, Y. Cui, Enhancing ionic conductivity in composite polymer electrolytes with well-aligned ceramic nanowires, Nat. Energy 2 (5) (2017) 17035, https://doi.org/10.1038/nenergy.2017.35.
[23] Z. Li, H.-M. Huang, J.-K. Zhu, J.F. Wu, H. Yang, L. Wei, X. Guo, Ionic conduction in composite polymer electrolytes: Case of PEO:Ga-LLZO composites, ACS Appl. Mater. Interfaces 11 (1) (2019) 784–791, https://doi.org/10.1021/acsami.8b17279.
[24] W. Wieczorek, M. Siekierski, A description of the temperature dependence of the conductivity for composite polymeric electrolytes by effective medium theory, J. Appl. Phys. 76 (4) (1994) 2220–2226, https://doi.org/10.1063/1.357638.
[25] F. Croce, G.B. Appetecchi, L. Persi, B. Scrosati, Nanocomposite polymer electrolytes for lithium batteries, Nature 394 (6692) (1998) 456–458, https://doi.org/10.1038/28818.
[26] A. Villa, J.C. Verduzco, J.A. Libera, E.E. Marinero, Ionic conductivity optimization of composite polymer electrolytes through filler particle chemical modification, Ionics 27 (6) (2021) 2483–2493, https://doi.org/10.1007/s11581-021-04042-9.
[27] S. Mogurampelly, V. Ganesan, Effect of nanoparticles on ion transport in polymer electrolytes, Macromolecules 48 (8) (2015) 2773–2786, https://doi.org/10.1021/ma502578s.
[28] S. Mogurampelly, V. Ganesan, Influence of nanoparticle surface chemistry on ion transport in polymer nanocomposite electrolytes, Solid State Ionics 286 (2016) 57–65, https://doi.org/10.1016/j.ssi.2015.12.018.
[29] M.R. Bonilla, F.A. García Daza, P. Ranque, F. Aguesse, J. Carrasco, E. Akhmatskaya, Unveiling interfacial Li-ion dynamics in $Li_7La_3Zr_2O_{12}$/PEO (LiTFSI) composite polymer-ceramic solid electrolytes for all-solid-state lithium batteries, ACS Appl. Mater. Interfaces 13 (26) (2021) 30653–30667, https://doi.org/10.1021/acsami.1c07029.
[30] D.J. Brooks, B.V. Merinov, W.A. Goddard, B. Kozinsky, J. Mailoa, Atomistic description of ionic diffusion in PEO-LiTFSI: Effect of temperature, molecular weight, and ionic concentration, Macromolecules 51 (21) (2018) 8987–8995, https://doi.org/10.1021/acs.macromol.8b01753.
[31] A. Ghoufi, G. Maurin, Microscopic model of the metal-organic framework/polymer interface: A first step toward understanding the compatibility in mixed matrix membranes, ACS Appl. Mater. Interfaces 8 (1) (2016) 809–819, https://doi.org/10.1021/acsami.5b10150.
[32] M. Marzantowicz, J. Dygas, F. Krok, A. Lasinska, Z. Florjanczyk, E. Zygadlo-Monikowska, A. Affek, Crystallization and melting of PEO:LiTFSI polymer electrolytes investigated simultaneously by impedance spectroscopy and polarizing microscopy, Electrochim. Acta 50 (19) (2005) 3969–3977, https://doi.org/10.1016/j.electacta.2005.02.053.
[33] W.L. Jorgensen, D.S. Maxwell, J. Tirado-Rives, Development and testing of the OPLS all-atom force field on conformational energetics and properties of organic liquids, J. Am. Chem. Soc. 118 (45) (1996) 11225–11236, https://doi.org/10.1021/ja9621760.
[34] L. Meabe, S.R. Peña, M. Martinez-Ibañez, Y. Zhang, E. Lobato, H. Manzano, M. Armand, J. Carrasco, H. Zhang, Insight into the ionic transport of solid polymer electrolytes in polyether and polyester blends, J. Phys. Chem. C 124 (33) (2020) 17981–17991, https://doi.org/10.1021/acs.jpcc.0c04987.
[35] T. Köddermann, D. Paschek, R. Ludwig, Molecular dynamic simulations of ionic liquids: A reliable description of structure, thermodynamics and dynamics, ChemPhysChem 8 (17) (2007) 2464–2470, https://doi.org/10.1002/cphc.200700552.
[36] T.C. Lim, Alignment of buckingham parameters to generalized Lennard-Jones potential functions, Zeitschrift für Naturforschung A 64 (3–4) (2009) 200–204, https://doi.org/10.1515/zna-2009-3-406.
[37] E. Akhmatskaya, S. Reich, GSHMC: An efficient method for molecular simulation, J. Comput. Phys. 227 (10) (2008) 4934–4954, https://doi.org/10.1016/j.jcp.2008.01.023.
[38] A. Horowitz, A generalized guided Monte Carlo algorithm, Phys. Lett. B 268 (2) (1991) 247–252, https://doi.org/10.1016/0370-2693(91)90812-5.
[39] B. Escribano, E. Akhmatskaya, J. Mujika, Combining stochastic and deterministic approaches within high efficiency molecular simulations, Central Eur. J. Math. 11 (4) (2013) 787–799, https://doi.org/10.2478/s11533-012-0164-x.
[40] M. Fernández-Pendás, E. Akhmatskaya, J. Sanz-Serna, Adaptive multi-stage integrators for optimal energy conservation in molecular simulations, J. Comput. Phys. 327 (2016) 434–449, https://doi.org/10.1016/j.jcp.2016.09.035.
[41] E. Akhmatskaya, M. Fernández-Pendás, T. Radivojević, J.M. Sanz-Serna, Adaptive splitting integrators for enhancing sampling efficiency of modified hamiltonian Monte Carlo methods in molecular simulation, Langmuir 33 (42) (2017) 11530–11542, https://doi.org/10.1021/acs.langmuir.7b01372.
[42] S. Pronk, S. Páll, R. Schulz, P. Larsson, P. Bjelkmar, R. Apostolov, M. Shirts, J. Smith, P. Kasson, D. Van Der Spoel, B. Hess, E. Lindahl, Gromacs 4.5: A high-throughput and highly parallel open source molecular simulation toolkit, Bioinformatics 29 (7) (2013) 845–854, https://doi.org/10.1093/bioinformatics/btt055.
[43] F.A. García Daza, M.R. Bonilla, A. Llordés, J. Carrasco, E. Akhmatskaya, Atomistic insight into ion transport and conductivity in ga/al-substituted $Li_7La_3Zr_2O_{12}$ solid electrolytes, ACS Appl. Mater. Interfaces 11 (1) (2019) 753–765, https://doi.org/10.1021/acsami.8b17217.
[44] M.R. Bonilla, F.A. García Daza, J. Carrasco, E. Akhmatskaya, Exploring Li-ion conductivity in cubic, tetragonal and mixed-phase al-substituted $Li_7La_3Zr_2O_{12}$ using atomistic simulations and effective medium theory, Acta Mater. 175 (2019) 426–435, https://doi.org/10.1016/j.actamat.2019.06.033.
[45] L. Martínez, R. Andrade, E.G. Birgin, J.M. Martínez, Packmol: A package for building initial configurations for molecular dynamics simulations, J. Comput. Phys. 30 (13) (2009) 2157–2164, https://doi.org/10.1002/jcc.21224.
[46] B. Hess, P-lincs: A parallel linear constraint solver for molecular simulation, J. Chem. Theory Comput. 4 (1) (2008) 116–122, https://doi.org/10.1021/ct700200b.
[47] J. Kästner, Umbrella sampling, WIREs Computational Molecular, Science 1 (6) (2011) 932–942, https://doi.org/10.1002/wcms.66.
[48] S. Kumar, J.M. Rosenberg, D. Bouzida, R.H. Swendsen, P.A. Kollman, The weighted histogram analysis method for free-energy calculations on biomolecules. i. the method, J. Comput. Chem. 13 (8) (1992) 1011–1021, https://doi.org/10.1002/jcc.540130812.
[49] R.C. Dutta, S.K. Bhatia, Structure and gas transport at the polymer-zeolite interface: Insights from molecular dynamics simulations, ACS Appl. Mater. Interfaces 10 (6) (2018) 5992–6005, https://doi.org/10.1021/acsami.7b17470.
[50] Y. Li, Y. Zhao, Y. Cui, Z. Zou, D. Wang, S. Shi, Screening polyethylene oxide-based composite polymer electrolytes via combining effective medium theory and Halpin-Tsai model, Comput. Mater. Sci. 144 (2018) 338–344, https://doi.org/10.1016/j.commatsci.2017.12.014.
[51] A. Maitra, A. Heuer, Cation transport in polymer electrolytes: A microscopic approach, Phys. Rev. Lett. 98 (2007) 227802, https://doi.org/10.1103/PhysRevLett.98.227802.
[52] S. Mogurampelly, O. Borodin, V. Ganesan, Computer simulations of ion transport in polymer electrolyte membranes, Ann. Rev. Chem. Biomol. Eng. 7 (1) (2016) 349–371, doi:10.1146/annurev-chembioeng-080615-034655.
[53] D. Roilo, P.N. Patil, R.S. Brusa, A. Miotello, S. Aghion, R. Ferragut, R. Checchetto, Polymer rigidification in graphene based nanocomposites: Gas barrier effects and free volume reduction, Polymer 121 (2017) 17–25, https://doi.org/10.1016/j.polymer.2017.06.010.
[54] S.G. Falkovich, V.M. Nazarychev, S.V. Larin, J.M. Kenny, S.V. Lyulin, Mechanical properties of a polymer at the interface structurally ordered by graphene, J. Phys. Chem. C 120 (12) (2016) 6771–6777, https://doi.org/10.1021/acs.jpcc.5b11028.
[55] P. Liu, E. Harder, B.J. Berne, On the calculation of diffusion coefficients in confined fluids and interfaces with an application to the liquid-vapor interface of water, J. Phys. Chem. B 108 (21) (2004) 6595–6602, https://doi.org/10.1021/jp0375057.
[56] K. Timachova, H. Watanabe, N.P. Balsara, Effect of molecular weight and salt concentration on ion transport and the transference number in polymer electrolytes, Macromolecules 48 (21) (2015) 7882–7888, https://doi.org/10.1021/acs.macromol.5b01724.
[57] H. Lee, A.H. de Vries, S.-J. Marrink, R.W. Pastor, A coarse-grained model for polyethylene oxide and polyethylene glycol: Conformation and hydrodynamics, J. Phys. Chem. B 113 (40) (2009) 13186–13194, https://doi.org/10.1021/jp9058966.
[58] B. Hanson, V. Pryamitsyn, V. Ganesan, Mechanisms underlying ionic mobilities in nanocomposite polymer electrolytes, *ACS Macro Lett.* 2(11) (2013) 1001–1005, doi: 10.1021/mz400234m.
[59] J. Zheng, M. Tang, Y.Y. Hu, Lithium ion pathway within $Li_7La_3Zr_2O_{12}$-polyethylene oxide composite electrolytes, Angew. Chem. Int. Ed. 55 (40) (2016) 12538–12542, https://doi.org/10.1002/anie.201607539.
[60] F. Croce, G.B. Appetecchi, L. Persi, B. Scrosati, Nanocomposite polymer electrolytes for lithium batteries, Nature 394 (6692) (1998) 456–458, https://doi.org/10.1038/28818.
[61] W. Wieczorek, K. Such, H. Wyciślik, J. Plocharski, Modifications of crystalline structure of peo polymer electrolytes with ceramic additives, Solid State Ionics 36 (3) (1989) 255–257, https://doi.org/10.1016/0167-2738(89)90185-9.
[62] C.-W. Nan, D.M. Smith, A.c. electrical properties of composite solid electrolytes, Mater. Sci. Eng. B 10 (2) (1991) 99–106, https://doi.org/10.1016/0921-5107(91)90115-C.